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CEE 406 Pavement Design I
Department of Civil and Environmental Engineering
University of Illinois at Urbana-Champaign
Homework #2
Due: 9/24/2024

1. Solve Problem 7.2, Huang, pg. 331. Please show your calculations. Refer to example 7.3 page 288 as practice.

2. The results from a resilient modulus test are as follows:

Confining Stress (kPa)	Deviator Stress (kPa)	Resilient Strain
20.68	18.41	0.000150
20.68	37.16	0.000270
20.68	55.85	0.000360
34.47	30.96	0.000190
34.47	62.05	0.000320
34.47	93.15	0.000440
68.95	61.91	0.000230
68.95	123.97	0.000440
68.95	186.50	0.000610
104.11	62.19	0.000200
103.42	93.08	0.000290
103.42	186.16	0.000510
137.90	93.08	0.000250
137.90	124.11	0.000320
137.90	248.42	0.000570
20.68	18.41	0.000190
20.68	37.30	0.000330
20.68	55.99	0.000440
34.47	31.03	0.000220
34.47	62.19	0.000390
34.47	93.42	0.000530
68.95	61.91	0.000270
68.95	124.59	0.000480
68.95	187.19	0.000660
103.42	62.19	0.000230
103.42	93.36	0.000320
103.42	186.50	0.000550
137.90	93.49	0.000270
137.90	124.45	0.000350

137.90	249.38	0.000600

Find the material parameters k_1, k_2, and k_3 in the equation and calculate R^2 (coefficient of determination). Please show your calculations.

$$M_r = k_1 p_a \left(\frac{\theta}{p_a}\right)^{k2} \left(\frac{\tau_{oct}}{p_a} + 1\right)^{k_3}$$

where $\theta = \sigma_1 + \sigma_2 + \sigma_3$ = first stress invariant; and

$$\tau_{oct} = \frac{1}{3}\sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2} \text{ = octahedral stresses}$$

3. Prepare a synthesis on subgrade stabilization using lime. It should include when to use lime, type of lime, and impacts of adding lime to soil (i.e., which kind of results/reactions would you expect when lime is added to soil). This part should not exceed one page.

4. In a CBR test, the following data were collected:

Penetration (in)	**Pressure on Piston (psi)**	
	Test Results	**Standard Stone Results**
0.05	10	---
0.1	30	1000
0.15	65	---
0.2	100	1500
0.3	150	1900
0.4	175	2300
0.5	200	2600

What is the CBR for this soil? Please show your calculations.